

A STUDY ON THE IMPACT OF COVID-19 ON DIGITAL PAYMENT BEHAVIOR AMONG CUSTOMERS IN INDIA

A Case Study of UPI

Ancilla Fiona DSouza

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Introduction

The digital payment system has become one of the most significant developments in India's financial sector in recent years. With rapid advances in technology, growing smartphone usage, wider internet access, and strong government support for a cashless economy, India has seen a remarkable surge in digital transactions.

The arrival of platforms like UPI, mobile wallets, internet banking, and QR-code payments has genuinely transformed how people manage their everyday finances. Then came COVID-19, and everything shifted even faster.

During the lockdown, people were urged to avoid physical contact and maintain social distance. Naturally, many turned away from cash and toward digital payments — not just out of convenience, but out of necessity. Digital transactions became safer, quicker, and simply more practical during those uncertain times.

Among all digital payment options, the Unified Payments Interface (UPI) stood out as the most widely adopted. Through apps like Google Pay, PhonePe, Paytm, and BHIM UPI, people could transfer money instantly from their phones without stepping into a bank. The platform's simplicity, speed, and security made it the go-to choice for millions of Indians during the pandemic.

The ripple effect was felt across the economy. Businesses, retailers, online platforms, and even small street vendors started accepting UPI to keep operations going during lockdown restrictions. Banks and fintech companies underwent rapid digital transformation as a result.

This report examines the impact of COVID-19 on digital payment behavior among customers in India, with a specific focus on UPI. The study draws on UPI transaction data from the National Payments Corporation of India (NPCI) covering the financial years 2019–20 and 2020–21 to understand how significantly behavior changed during the pandemic.

Objectives of the Study

- ☐ To study the impact of COVID-19 on digital payment behavior among customers in India.
- ☐ To analyze the growth of UPI transactions during the pre-COVID and COVID periods.
- ☐ To compare UPI transaction volume and transaction value between 2019–20 and 2020–21.
- ☐ To examine the increase in customer adoption of digital payment systems during the pandemic.
- ☐ To analyze trends in bank-level participation and digital transaction growth.
- ☐ To understand the role of UPI in promoting cashless transactions during COVID-19

Scope of the Study

The study focuses on analyzing the impact of COVID-19 on digital payment behavior in India with special reference to UPI transactions. The research is limited to the financial years 2019–20 and 2020–21 to compare the pre-COVID and COVID periods.

The study uses secondary data collected from the official reports published by the National Payments Corporation of India (NPCI). The analysis mainly covers:

- Transaction volume
- Transaction value
- Growth percentage
- Bank-level participation
- Trend analysis through graphical representation

The study helps in understanding how customers shifted from cash payments to digital transactions during the pandemic period.

Limitations of the Study

- ☐ The study is based only on secondary data collected from NPCI reports and published sources.
- ☐ The study is limited to UPI transactions and does not cover all digital payment systems.
- ☐ The analysis covers only the financial years 2019–20 and 2020–21.
- ☐ Changes in customer behavior are interpreted mainly through transaction statistics and not through direct surveys or interviews.
- ☐ The study may not fully represent digital payment behavior in all regions of India, especially remote rural areas.

Industry Profile of UPI

Unified Payments Interface (UPI) is a real-time digital payment system built and managed by the National Payments Corporation of India (NPCI). It enables users to transfer money instantly between bank accounts through a mobile app — without needing to share detailed banking information every time.

UPI was created to build a fast, secure, and user-friendly payment ecosystem in India. It links multiple bank accounts to a single mobile application and allows users to carry out transactions around the clock, every day of the year. From sending money to a friend, to paying a utility bill, recharging a phone, or shopping online — it all happens in seconds.

NPCI plays a central role in managing the UPI infrastructure and ensuring that transactions across banks and fintech platforms remain smooth and secure. The platform has been widely embraced because of how accessible and straightforward it is.

Apps like Google Pay, PhonePe, Paytm, Amazon Pay, and BHIM UPI have brought UPI to both urban centers and rural communities, making digital payments genuinely inclusive. The COVID-19 pandemic only added fuel to this growth — as people sought contactless ways to pay, UPI became indispensable for individuals, businesses, and merchants alike.

Literature Review

Impact of COVID-19 on Digital Payment Usage in India

Authors: Dr. Rashmi Mate and Dr. Atul Kapdi

This study looked at how COVID-19 changed the way people used digital payments across India. The authors found that lockdowns and social distancing pushed a large number of people away from cash and toward digital alternatives. Apps like Paytm, BHIM UPI, Google Pay, and PhonePe became part of daily life, offering safe and contactless ways to pay.

The research used both primary and secondary data — primary data came from 113 survey respondents via Google Forms, while secondary data was gathered from RBI reports and government portals. The study found a clear increase in e-wallet usage during COVID-19, with most users citing ease of use, time-saving, and safety as the main reasons for switching.

The authors also noted that the Reserve Bank of India actively promoted digital payment systems like UPI, NEFT, IMPS, and internet banking to reduce physical interactions. Their conclusion was that COVID-19 served as a powerful catalyst, pushing India closer to becoming a cashless economy.

Impact of COVID-19 on Digital Payment Services at Towns and Villages

Authors: Ms. Rashmi Singh and Abhilasha Gupta

This study focused specifically on how digital payments changed in smaller towns and villages — areas that had previously seen limited adoption of digital tools. The authors found that COVID-19 significantly raised awareness about digital payments even in rural India, with small shopkeepers, vendors, and everyday consumers beginning to use apps for transactions.

The research relied primarily on secondary data from journals, articles, and government reports. The authors reviewed existing literature on topics like demonetization, Digital India initiatives, and previous efforts to bring cashless payments to rural communities.

Their findings were encouraging — rural digital payment usage went up considerably during the pandemic. However, the study also acknowledged persistent challenges: limited digital literacy, patchy internet connectivity, and a general lack of awareness in remote areas still stood in the way of full adoption.

Product Statistics of UPI (2019–2020)

The following table presents the NPCI product statistics for UPI transactions during the financial year 2019–20. This data represents the pre-COVID baseline used for comparative analysis.

Month	No. of Banks live on UPI	Volume (In Mn.)	Value (In Cr.)	Period	Vol Growth %	Value Growth %
1-Apr-19	144	781.79	142034.39	Pre-COVID	NA	NA
1-May-19	143	733.54	152449.29	Pre-COVID	-6.172	7.333
1-Jun-19	142	754.54	146566.35	Pre-COVID	2.863	-3.859
1-Jul-19	143	822.29	146386.64	Pre-COVID	8.979	-0.123
1-Aug-19	141	918.35	154504.89	Pre-COVID	11.682	5.546
1-Sep-19	141	955.02	161456.56	Pre-COVID	3.993	4.499
1-Oct-19	141	1148.36	191359.94	Pre-COVID	20.245	18.521
1-Nov-19	143	1218.77	189229.09	Pre-COVID	6.131	-1.114
1-Dec-19	143	1308.4	202520.76	Pre-COVID	7.354	7.024
1-Jan-20	144	1305.02	216242.97	Pre-COVID	-0.258	6.776
1-Feb-20	146	1325.69	222516.95	Pre-COVID	1.584	2.901
1-Mar-20	148	1246.84	206462.31	Pre-COVID	-5.948	-7.215

Product Statistics of UPI (2020–2021)

The table below captures the NPCI product statistics for UPI transactions during the financial year 2020–21, which largely covers the COVID-19 pandemic period and reflects the sharp changes in digital payment behavior.

Month	No. of Banks live on	Volume (In Mn.)	Value (In Cr.)	Period	Vol Growth %	Value Growth %
1-Apr-20	153	999.57	151140.66	COVID	NA	NA
1-May-20	155	1234.5	218391.6	COVID	23.503	44.496
1-Jun-20	155	1336.93	261835	COVID	8.297	19.892
1-Jul-20	164	1497.36	290537.86	COVID	12.000	10.962
1-Aug-20	168	1618.83	298307.61	COVID	8.112	2.674
1-Sep-20	174	1800.14	329027.66	COVID	11.200	10.298
1-Oct-20	189	2071.62	386106.74	COVID	15.081	17.348
1-Nov-20	200	2210.23	390999.15	COVID	6.691	1.267
1-Dec-20	207	2234.16	416176.21	COVID	1.083	6.439
1-Jan-21	207	2302.73	431181.89	COVID	3.069	3.606
1-Feb-21	213	2292.9	425062.76	COVID	-0.427	-1.419
1-Mar-21	216	2731.68	504886.44	COVID	19.136	18.779

Research Methodology

This study is descriptive and analytical in nature. The aim is to understand how COVID-19 influenced digital payment behavior among customers in India, with a particular focus on UPI transactions.

The study is entirely based on secondary data, collected from official statistical reports published by NPCI. Additional material was drawn from research journals, RBI reports, government publications, and credible online sources related to digital payments.

The study covers the financial years 2019–20 and 2020–21, deliberately chosen to allow a direct comparison between the pre-COVID and COVID periods.

Microsoft Excel was used to organize and analyses the data. The following analytical tools and techniques were applied:

- Product Statistics Analysis
- Comparative Analysis
- Average Bank-Level Analysis

- Average Transaction Volume Analysis
- Average Transaction Value Analysis
- Volume Growth Percentage
- Value Growth Percentage
- Trend Analysis
- Graphical Representation

Visual aids — including volume trend graphs, value trend graphs, bank trend graphs, and pre-COVID vs. COVID comparison charts — were prepared to illustrate the growth patterns in UPI usage during the pandemic.

Data Analysis and Interpretation

The study analyzed UPI transaction statistics for 2019–20 and 2020–21 using NPCI data on transaction volume and value.

The data tells a clear story: UPI usage grew consistently and significantly during COVID-19. Customers turned to digital payments because they were convenient, fast, and crucially — safe. During lockdowns, digital transactions were not just preferred; they were often the only practical option for buying essentials, sending money, or paying bills.

The average transaction volume in 2020–21 was notably higher than in 2019–20, indicating that more customers were using UPI apps regularly for daily financial needs. The average transaction value also rose during the same period, reflecting a broader and deeper reliance on digital financial services.

Comparative Analysis

Comparing 2019–20 with 2020–21 reveals a striking acceleration in UPI activity during the COVID-19 period.

The surge in transaction volume was driven by a clear shift away from cash. Online shopping, utility bill payments, mobile recharges, and fund transfers all increased as people relied more heavily on their phones for financial tasks. The transaction value also grew considerably, showing that UPI was being used not just for small purchases but for increasingly significant financial activities.

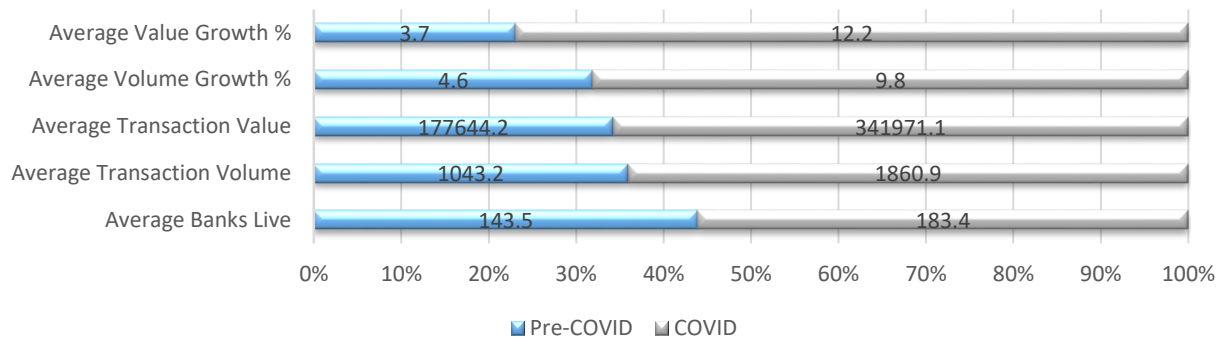
Indicator	Pre-COVID	COVID
Average Banks Live	143.5	183.4
Average Transaction Volume	1043.2	1860.9
Average Transaction Value	177644.2	341971.1
Average Volume Growth %	4.6	9.8
Average Value Growth %	3.7	12.2

Graphical Representation and Trend Analysis

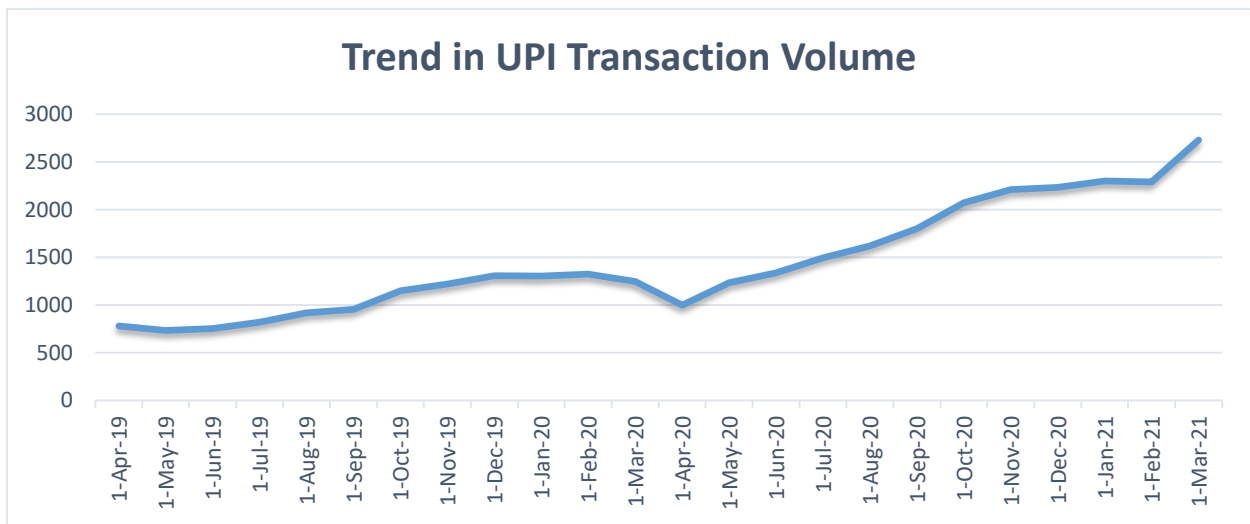
Graphical analysis was used to visualize growth trends and customer behavior in UPI transactions during COVID-19. The charts consistently show an upward trajectory in transaction volume, value, and bank-level participation.

Pre-COVID vs. COVID UPI Comparison Graph

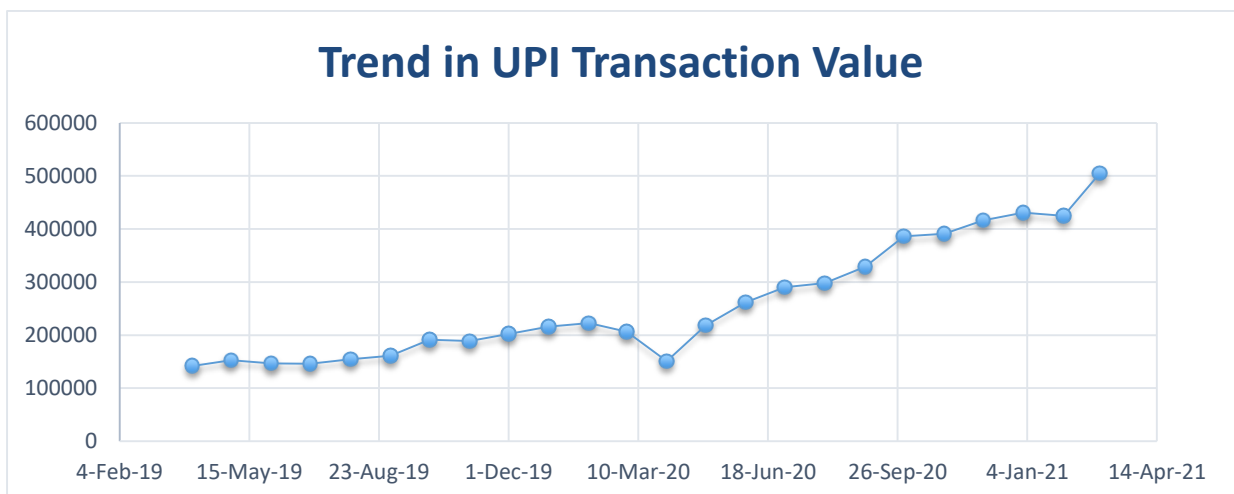
Pre-COVID vs COVID UPI Comparison



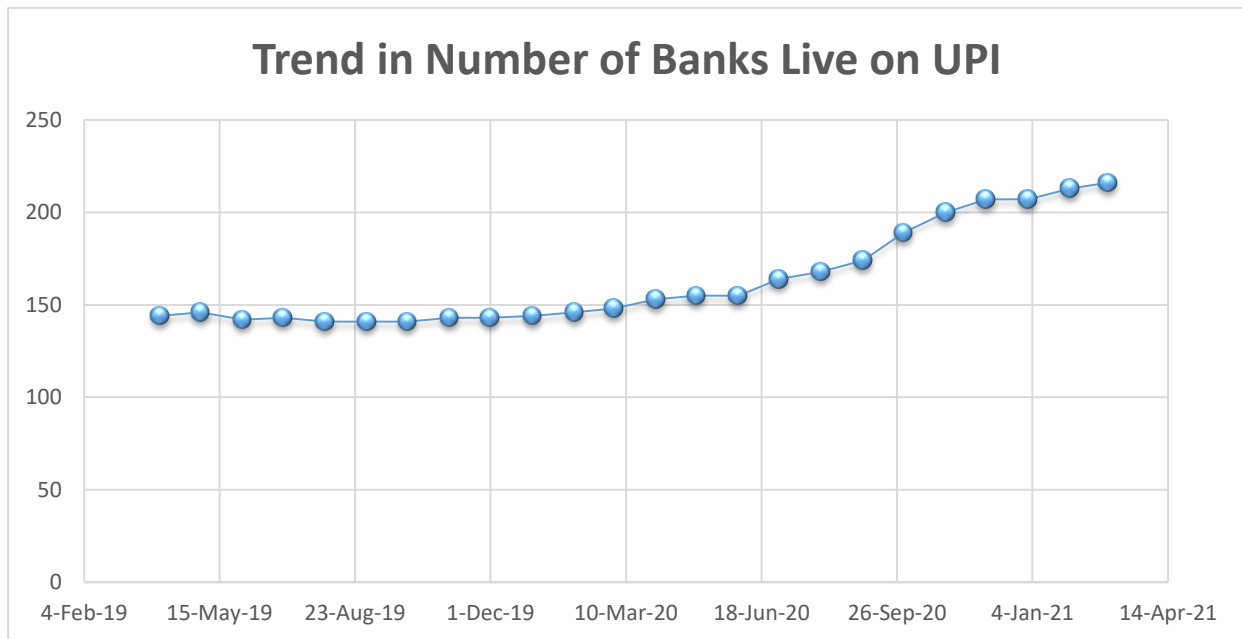
Volume Trend Graph



Value Trend Graph



Bank Trend Graph



Taken together, these visuals confirm what the data already suggests: COVID-19 accelerated India's transition to digital payments in a way that would have taken years to achieve otherwise.

The volume trend graph highlights steady month-on-month growth in the number of digital transactions, fueled by increased reliance on UPI during lockdowns. The value trend graph shows a corresponding rise in the total monetary value of UPI transactions — people were not just transacting more often; they were transacting larger amounts. The bank trend graph demonstrates growing participation from banks and fintech companies in the UPI ecosystem, signaling widespread institutional support for digital payments.

Impact on Commerce and Finance

The growth of UPI during COVID-19 had a tangible impact on both commerce and finance in India.

On the commerce side, digital payments helped businesses survive lockdown restrictions. Online shopping platforms, grocery stores, pharmacies, and local vendors — many of whom had never used digital payments before — quickly adopted UPI to keep serving customers. QR-code-based transactions became particularly popular, even among small traders and roadside vendors who previously operated exclusively in cash.

In the financial sector, UPI significantly improved the efficiency and accessibility of transactions. Customers could transfer money instantly without visiting a bank branch, and the banking sector underwent rapid digital transformation in response to growing demand. Mobile banking and online payment services became mainstream almost overnight.

UPI also played an important role in advancing financial inclusion. By making digital banking services available to people in both cities and villages, it brought more Indians into the formal financial system. The fintech industry benefited enormously — apps that had already been growing quickly saw explosive user growth as more people embraced digital payments out of necessity.

Challenges and Risks

Despite the impressive growth, the study identified several challenges and risks that remain relevant:

- Cyber fraud and online scams targeting digital payment users
- Data privacy and security concerns around financial information
- Poor internet connectivity in rural and semi-urban areas

- Low digital literacy, particularly among older adults and rural populations
- Technical server failures and service disruptions
- Heavy dependence on smartphones and reliable internet access

These challenges highlight the need for stronger cybersecurity measures, better digital infrastructure, and sustained awareness programmes — particularly in areas where digital adoption is still growing.

Findings

- COVID-19 significantly accelerated digital payment adoption among customers in India.
- UPI transaction volume and value both recorded strong growth during 2020–21.
- Customers moved away from cash toward contactless digital payments primarily for health and safety reasons.
- Online shopping and digital financial activities increased substantially during the pandemic.
- UPI became the most preferred digital payment method across age groups and geographies.
- Banks and fintech companies recorded higher digital transaction growth during this period.
- Small businesses and local vendors adopted QR-code-based payments to continue operating during lockdowns.
- The pandemic contributed meaningfully to India's progress toward a cashless economy.

Suggestions

- Strengthen cybersecurity frameworks to better protect users from fraud and scams.
- Run targeted awareness campaigns on safe digital payment practices, especially in rural areas.
- Invest in digital literacy programmes for older adults and first-time digital payment users.
- Expand internet connectivity and improve digital infrastructure in underserved areas.
- Develop more robust technical support systems to reduce UPI service disruptions.
- Incentivize small businesses and merchants to adopt and maintain digital payment technologies.

Conclusion

This study confirms that COVID-19 had a profound and lasting impact on digital payment behavior among customers in India. The pandemic did not simply speed up an existing trend — it fundamentally changed how millions of people interact with money.

The data from NPCI for 2019–20 and 2020–21 clearly shows substantial growth in both the volume and value of UPI transactions during the COVID-19 period. Customers, businesses, banks, and fintech companies all increased their reliance on UPI for financial transactions, often out of necessity but staying for the convenience.

UPI proved its worth as a financial lifeline during lockdown restrictions. It kept commerce moving, enabled financial inclusion, and supported India's broader digital transformation. Yes, challenges remain — cybersecurity risks, digital literacy gaps, and uneven internet access are real concerns that need ongoing attention. But the progress made is undeniable.

In many ways, COVID-19 compressed years of potential growth into just a few months. India's digital payment ecosystem came out of the pandemic stronger, more inclusive, and far more deeply embedded in everyday life than it had been before.

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